⚡ **Chemistry 30 | Thermochemical Changes | Lesson 8**

Empirical Effects of Catalysis

catalysts • energy barriers • reaction pathways • enzymes

🧠 Big Idea

A catalyst does not “create” energy and it does not change how much energy a reaction releases or absorbs. What it changes is the path the reaction can take.

Catalysts speed reactions up by providing an alternate pathway with a lower activation energy (E_a). The reactants and products are the same, so the overall enthalpy change (ΔH) is the same.

1) What catalysts do: evidence first, then explanation

In the Thermochemistry unit, you have learned to describe energy changes using ΔH and energy diagrams. Today we add one more piece: even when a reaction is energetically possible, it might still be too slow to notice. If you have ever wondered why gasoline in a can does not “explode” on its own, or why food spoils more slowly in the refrigerator, you have already met the idea behind activation energy. A reaction can be downhill overall (negative ΔH) and still require an initial push to get started.

A catalyst is a substance that increases the rate of a reaction without being permanently consumed. The most important word here is “rate.” When a catalyst is present, you observe that the reaction happens faster: gas bubbles form more quickly, temperature changes occur sooner, colour changes happen rapidly, or a measurable amount of product appears in less time. What you do not observe is a different set of products or a different overall energy change for the same reaction.

A classic empirical example is the decomposition of hydrogen peroxide: $2\ H_2O_2(aq) \rightarrow 2\ H_2O(l) + O_2(g)$. On its own, hydrogen peroxide decomposes slowly at room temperature. If you add a small amount of manganese(IV) oxide, $MnO_2(s)$, the reaction becomes dramatically faster and oxygen gas is produced quickly. If the same amount of hydrogen peroxide is used each time, the total amount of oxygen that can be produced (the chemical amount, in moles) does not change. The catalyst changes the time required to produce oxygen, not the maximum amount of oxygen possible from the limiting reactant.

To explain those observations, chemists use the energy-barrier model. For particles to react, they must collide with enough energy and in a suitable orientation to reach the transition state (the highest-energy point along the reaction path). The minimum energy needed to reach that point is the activation energy, E_a. A catalyst provides an alternate pathway—often a different mechanism made of several smaller steps—in which the highest point is lower in energy. Because the peak is lower, a larger fraction of collisions have enough energy to react at the same temperature, so the reaction rate increases.

Here is the comparison to keep straight: ΔH describes the energy difference between products and reactants (final minus initial). E_a describes the energy required to climb from the reactant level up to the peak (the barrier). A catalyst lowers E_a, but it does not move the starting level (reactants) or the ending level (products). That is why an energy diagram with a catalyst shows the same reactant and product energies, but a smaller hump. The diagram is deliberately separating "how much energy is transferred overall" from "how hard it is to get the reaction started."

Finally, catalysts do not change the net energy involved because ΔH is a state function. It depends only on the initial and final states, not on the route taken between them. A catalyst changes the route. It does not change where you start or where you finish. This is also why catalysts do not change the position of equilibrium: they speed up both the forward and reverse reactions, helping the system reach equilibrium faster, but not changing the equilibrium mixture itself. In living systems, enzymes act as catalysts by binding reactants in very specific ways and stabilizing the transition state, which lowers E_a while keeping ΔH unchanged.

📘 Key Vocabulary

Catalyst: a substance that increases reaction rate by providing an alternate pathway, and is regenerated at the end of the reaction.

Activation energy (E_a): the minimum energy required for reacting particles to reach the transition state and form products.

Alternate pathway / mechanism: a different sequence of steps that leads from reactants to products with a lower energy barrier.

Enzyme: a biological catalyst (usually a protein) that speeds up reactions in living systems by lowering E_a in a highly specific way.

⚠️ Common Error

Mixing up E_a and ΔH. Students sometimes assume that "lower energy barrier" means "more energy released." These are different ideas: E_a affects speed; ΔH describes overall energy change.

Another common mistake is saying a catalyst "makes more product." A catalyst can make products appear faster, but it does not change the limiting reagent, the balanced equation, or the final equilibrium amounts.

🎓 Diploma Tip

On an energy diagram, label ΔH as the vertical difference between reactants and products. Label E_a as the vertical difference between reactants and the peak.

If a catalyst is added, redraw the peak lower but keep the reactant and product levels unchanged. If you move the start or end levels, you have accidentally changed ΔH.

⏸ Pause & Think

A student runs the hydrogen peroxide decomposition reaction twice. Trial 1 uses 50.0 mL of $H_2O_2(aq)$ and no catalyst. Trial 2 uses the same amount of $H_2O_2(aq)$ but adds a small amount of $MnO_2(s)$. In Trial 2, oxygen is produced much faster.

Explain, in complete sentences, what is different in the energy diagram for Trial 2 compared to Trial 1. Your answer must clearly mention E_a and ΔH and state what changes and what stays the same.

🧠 Big Idea

Catalysts speed reactions up by lowering the activation-energy barrier (E_a), not by changing how much energy the reaction releases or absorbs (ΔH). On an energy diagram, that means the *peak* changes, but the *start* and *finish* do not. If you can keep that one picture in your head, most catalyst questions become straightforward.

2) Reading a catalyzed energy diagram (E_a changes, ΔH does not)

Energy diagrams are a way of telling the “energy story” of a reaction. The vertical axis is potential energy (often in $kJ\cdot mol^{-1}$), and the horizontal axis is reaction progress (sometimes called reaction coordinate). The reaction coordinate is not time—it is a pathway from reactants to products.

The first thing to do on any energy diagram is anchor your eyes on the endpoints. The left-hand level is the potential energy of the reactants; the right-hand level is the potential energy of the products. The vertical difference between those two levels is the enthalpy change, ΔH. If products are lower, ΔH is negative (exothermic). If products are higher, ΔH is positive (endothermic).

Next, find the highest point on the curve. That peak represents the transition state—an unstable arrangement that must form briefly as old bonds are stretched and broken and new bonds begin to form. The activation energy, E_a, is the vertical energy difference from the reactant level up to the peak. This is the “energy barrier” that limits rate.

Now add a catalyst. A catalyst provides an alternate reaction pathway (often through a different mechanism with different intermediates). The key idea is that the catalyst changes the *route*, not the *destination*. Reactants are still reactants, and products are still products—so the endpoint energies (and therefore ΔH) stay the same. What changes is the height of the barrier, because the catalyzed pathway has a lower-energy transition state (or a series of smaller barriers).

Worked example (numbers): Imagine an exothermic reaction where reactants are at 0 $kJ\cdot mol^{-1}$ and products are at -50 $kJ\cdot mol^{-1}$. Without a catalyst, the peak is at $+120$ $kJ\cdot mol^{-1}$. With a catalyst, the peak is at $+70$ $kJ\cdot mol^{-1}$. The enthalpy change is $\Delta H = -50$ $kJ\cdot mol^{-1}$ in both cases because the start and finish levels are unchanged. But the activation energies are different: E_a(uncat) = 120 $kJ\cdot mol^{-1}$ and E_a(cat) = 70 $kJ\cdot mol^{-1}$. Lower barrier → more successful collisions per second → faster rate.

A common point of confusion is to think "lower peak means less energy involved." The peak is not the total energy change. It is the barrier height. The total energy change is still the difference between reactants and products. Catalysts do not change ΔH because ΔH depends only on initial and final states (a state function), not on the pathway taken. That path-independence is the same reasoning that makes Hess' Law work.

Finally, remember that many catalyzed reactions happen in multiple steps. On a multi-step energy diagram you may see several peaks. The highest peak (largest barrier from the previous valley) is the rate-determining step. A catalyst may lower one or more peaks, often changing which step is rate-determining, but the overall ΔH (reactants to products) still remains unchanged.

📘 Key Vocabulary

Activation energy (E_a): the minimum energy barrier the reactants must overcome to reach the transition state.
Transition state: the highest-energy arrangement along a reaction pathway; it is not an isolatable substance.
Reaction coordinate: a "path" variable that tracks progress from reactants to products (not time).
Mechanism: the step-by-step sequence of elementary reactions that adds up to the overall reaction.
Intermediate: a species formed in one step and used up in a later step; it does not appear in the net reaction.
Rate-determining step: the slowest step (largest barrier) that limits the overall reaction rate.
Catalyst: a substance that increases rate by providing an alternate pathway and is regenerated.
Enzyme: a biological catalyst; an active site binds reactants and lowers E_a in a highly specific way.

⚠️ Common Error

Assuming a catalyst "makes the reaction more exothermic." It does not. A catalyst changes E_a (the barrier) but does not change ΔH (the net energy change).

Another frequent mistake is labeling E_a from the products up to the peak. By definition, E_a is measured from the reactants up to the transition state for the forward reaction (and from products up to the peak for the reverse reaction).

🎓 Diploma Tip

When you are asked to sketch or interpret an energy diagram with and without a catalyst:
1) Keep reactant and product energy levels identical on both diagrams.
2) Draw the catalyzed curve with a lower peak (or multiple smaller peaks).
3) Label ΔH once as the vertical difference between reactants and products.
4) Label E_a separately for the catalyzed and uncatalyzed pathways.
If you write one sentence of justification, make it: "ΔH depends only on initial and final states, so changing the pathway (catalyst) cannot change ΔH."

⏸ Pause & Think

You are given an energy diagram for an exothermic reaction. The reactants are at 10 $kJ·mol^{-1}$, the products are at −40 $kJ·mol^{-1}$, and the peak is at 90 $kJ·mol^{-1}$.
a) Calculate ΔH and E_a for the forward reaction.
b) A catalyst is added and the peak drops to 55 $kJ·mol^{-1}$. What changes? What must stay the same?
c) In one sentence, explain why the catalyst cannot change ΔH even though the reaction now happens faster.

🧠 Big Idea

The clearest proof that catalysts “work” is not a definition—it is a measurement. When the reactants and conditions stay the same, adding a catalyst makes the reaction reach the same end point in less time. That is the empirical signature of catalysis: the pathway changes (lower Ea), but the start and end energies (ΔH) do not.

3) Empirical evidence of catalysis: what you can actually measure

At this point you can say the words “a catalyst lowers activation energy,” but chemistry students earn full marks when they can explain how we *know* that claim is true. In other words: if you walked into a lab with no textbook, what observations would convince you that a catalyst changed the reaction pathway?

Start with the idea of reaction rate. Rate is not “how much energy” and it is not “how much product.” Rate is a *how fast* idea. In a lab, you measure rate by tracking a change you can observe over time—such as the volume of gas produced, the mass of a reactant remaining, the intensity of a colour, or the time it takes for a visible change (like cloudiness or bubbling) to occur.

Now consider a classic demonstration: the decomposition of hydrogen peroxide, $H_2O_2(aq)$, into water and oxygen gas. Without a catalyst, the reaction is so slow that you might barely notice anything happening. If you add a small amount of $MnO_2(s)$ or KI(aq), oxygen bubbles appear rapidly and the reaction finishes much sooner. The key is that the *products are still the same* (H_2O and O_2), and if you start with the same amount of H_2O_2, you end with essentially the same total amount of O_2—just produced faster.

This is exactly what a catalyst is allowed to change. A catalyst changes the *rate law pathway* (the step-by-step route particles follow), which changes the activation energy barrier. It does not create extra energy, and it does not change the energy difference between reactants and products. That is why energy diagrams show the peak (Ea) lowering while the vertical difference between reactants and products (ΔH) stays the same.

To make the evidence convincing, you must control variables. If you run a “with catalyst” trial at a higher temperature or with better stirring, you will not know what caused the rate increase. Good experimental thinking keeps temperature, volume, concentration, surface area (when relevant), and mixing conditions consistent—and changes only one factor: the presence (or amount) of catalyst.

Finally, notice what happens to the catalyst itself. In many demonstrations, the catalyst can be recovered at the end (for example, MnO_2 is still present). Even when a catalyst changes form during the mechanism, it is regenerated by the end of the reaction. That is why we say catalysts participate in the pathway but are not consumed overall.

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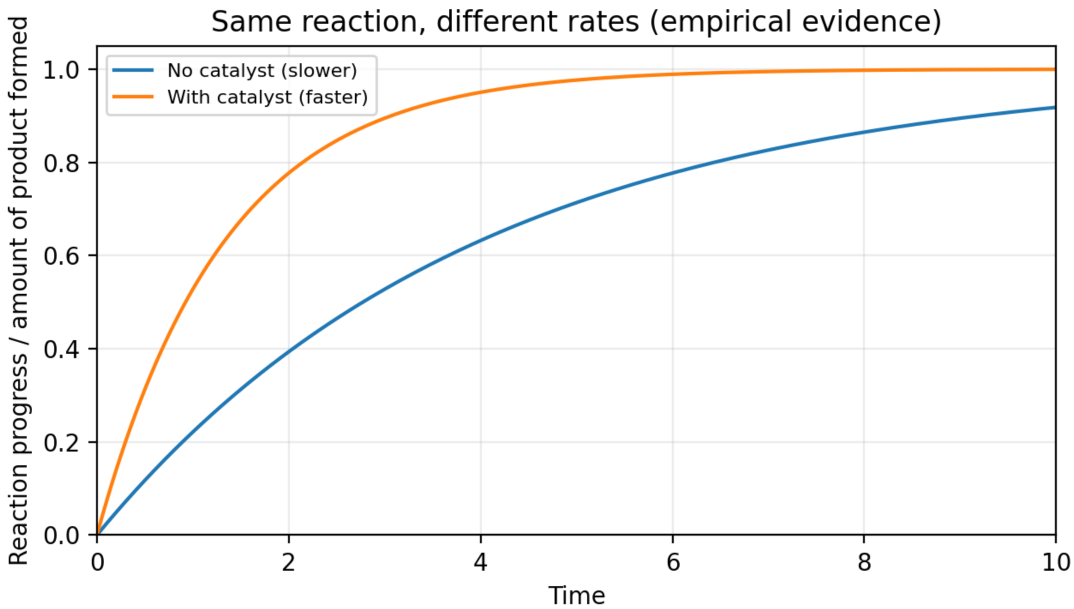
Worked example (numbers): Imagine an exothermic reaction where reactants are at 0 $kJ \cdot mol^{-1}$ and products are at −50 $kJ \cdot mol^{-1}$. Without a catalyst, the peak is at +120 $kJ \cdot mol^{-1}$. With a catalyst, the peak is at +70 $kJ \cdot mol^{-1}$. The enthalpy change is $\Delta H = -50\ kJ \cdot mol^{-1}$ in both cases because the start and finish levels are unchanged. But the activation energies are different: $E_a(\text{uncat}) = 120\ kJ \cdot mol^{-1}$ and $E_a(\text{cat}) = 70\ kJ \cdot mol^{-1}$. Lower barrier → more successful collisions per second → faster rate.

A common point of confusion is to think "lower peak means less energy involved." The peak is not the total energy change. It is the barrier height. The total energy change is still the difference between reactants and products. Catalysts do not change ΔH because ΔH depends only on initial and final states (a state function), not on the pathway taken. That path-independence is the same reasoning that makes Hess' Law work.

Finally, remember that many catalyzed reactions happen in multiple steps. On a multi-step energy diagram you may see several peaks. The highest peak (largest barrier from the previous valley) is the rate-determining step. A catalyst may lower one or more peaks, often changing which step is rate-determining, but the overall ΔH (reactants to products) still remains unchanged.



📘 Key Vocabulary

Reaction rate: how quickly reactants are consumed or products are formed (always tied to a time interval).
Catalyst: a substance that increases reaction rate by providing an alternative pathway; it is regenerated by the end.
Control variable: a condition kept the same in all trials (temperature, concentration, volume, stirring, surface area, etc.).
Independent variable: the factor you intentionally change (here, the presence/amount/type of catalyst).
Dependent variable: what you measure (gas volume vs time, time to completion, temperature change vs time, etc.).
Homogeneous catalyst: in the same phase as reactants (often all aqueous or all gaseous).
Heterogeneous catalyst: in a different phase (often a solid catalyst with liquid/gas reactants).
Enzyme: a biological catalyst—usually a protein—that speeds up reactions in living systems.

⚠️ Common Error

Confusing “faster” with “more energy.” A catalyst can make a reaction happen quickly even if ΔH is small, and it cannot make ΔH bigger or smaller.
Changing more than one variable at once (e.g., adding a catalyst *and* heating the mixture), then claiming the catalyst caused the rate change.
Treating the catalyst as a reactant that gets used up. In an overall equation, the catalyst should cancel out because it is regenerated.
Looking only at bubbles or excitement and ignoring the key evidence: the same products and the same final amount—just reached sooner.

🎓 Diploma Tip

In diploma-style explanations, build the argument in this order: (1) define the system (reaction) and what “rate” means, (2) describe the evidence as a measurement over time (a steeper slope or shorter time to reach the same end point), (3) connect the evidence to energy diagrams: Ea decreases while ΔH stays constant, and (4) state that the catalyst is regenerated, so it changes the pathway, not the overall chemical change.

⏸ Pause & Think

1) A graph shows two trials of the same reaction: Trial A reaches a plateau in 12 minutes, Trial B reaches the same plateau in 4 minutes. What does this tell you about (a) reaction rate, (b) ΔH, and (c) activation energy? Explain briefly.

2) Design a simple fair test to investigate whether a catalyst speeds up the decomposition of H_2O_2. List one independent variable, two control variables, and one dependent variable you would measure.

🧠 Big Idea

Enzymes are nature’s catalysts. They make the chemistry of life fast enough to happen at body temperature by lowering the activation energy (Ea), while leaving the overall energy change (ΔH) and the final equilibrium unchanged.

4) Enzymes as catalysts in living systems: speed with control

Up to now, we have treated a catalyst as “anything that speeds up a reaction without being consumed.” That definition is correct—but it feels abstract until you look at enzymes. Enzymes are catalysts made by living organisms, and they are the reason your body can run thousands of chemical reactions every second at about 37°C. Without enzymes, many of those reactions would still be thermodynamically possible (they might have a negative ΔH, or be favourable overall), but they would occur so slowly that life could not function.

Here is the key idea: an enzyme does not “add energy” to a reaction. Instead, it changes the pathway. The enzyme provides an alternate route from reactants to products that has a smaller activation energy barrier. On an energy diagram, the reactants and products sit at the same potential energies with or without the enzyme, so ΔH is the same. The only change is the height of the hump (Ea).

How does an enzyme lower Ea? Enzymes have an active site—a small region whose shape and chemical environment are a good fit for specific reactant molecules, called substrates. When substrates bind to the active site, several helpful things can happen at once: the enzyme can hold reacting particles in the correct orientation, it can slightly strain bonds that need to break, and it can create a micro-environment (local charges or polarity) that stabilizes the high-energy transition state. All of these effects lower the energy needed to reach the transition state, which is what “lowering Ea” means in physical terms.

A common point of confusion is whether enzymes change how much product is produced. They do not. Because a catalyst lowers Ea for both the forward and the reverse reaction, equilibrium is reached faster, but the equilibrium position (the ratio of products to reactants at equilibrium) does not change. Think of it this way: a catalyst does not change the start and end points on the energy scale; it only builds a better road between them.

Enzymes also help us see why catalysts can “stop working.” Enzymes are proteins with a specific three-dimensional shape. If temperature is too high or pH is far from the enzyme’s optimum, the enzyme can denature (lose its shape). When the shape changes, substrates no longer fit the active site well, and the catalytic effect drops sharply—even though the reaction itself still has the same ΔH.

Real-world example: catalase is an enzyme in your cells that speeds up the decomposition of hydrogen peroxide, $H_2O_2(aq)$, into water and oxygen. Hydrogen peroxide forms in small amounts in cells and can damage tissues, so catalase must remove it quickly. In a lab demonstration, adding a small amount of catalase (from yeast or liver) causes immediate bubbling of O_2—clear empirical evidence of a higher reaction rate. The energy released by the reaction is the same; it is simply released faster.

Checkpoint reasoning you should be able to explain: "The enzyme increases the rate because it lowers Ea by stabilizing the transition state. The overall ΔH is unchanged because the reactants and products are unchanged. The catalyst affects kinetics, not the thermodynamic energy difference between starting and ending states."

📘 Key Vocabulary

Enzyme: a biological catalyst (usually a protein) that speeds up a specific reaction.
Substrate: the reactant molecule(s) an enzyme acts on.
Active site: the region of an enzyme where substrates bind and the reaction is catalyzed.
Transition state: a short-lived, high-energy arrangement of atoms at the top of the activation barrier.
Denaturation: loss of an enzyme's specific shape (and function) due to temperature, pH, or chemicals.
Inhibitor: a substance that reduces enzyme activity by blocking the active site or altering its shape.

⚠️ Common Error

Thinking that enzymes "change ΔH" or "make more product." Enzymes lower Ea (the barrier), not the energy difference between reactants and products. They speed up both forward and reverse reactions equally, so equilibrium is reached sooner, but the equilibrium position and ΔH stay the same.

🎓 Diploma Tip

When you write a diploma explanation about enzymes/catalysts, use this structure: (1) state that the catalyst increases reaction rate; (2) explain that it provides an alternate pathway with lower Ea; (3) point to an energy diagram: Ea decreases, ΔH stays the same; (4) conclude that catalysts affect kinetics (speed) but not the overall energy change or equilibrium position.

⏸ Pause & Think

1) Sketch or interpret an energy diagram for a reaction with and without an enzyme. Label reactants, products, ΔH, Ea (uncatalyzed), and Ea (catalyzed).

2) An enzyme-catalyzed reaction runs fastest at 37°C. Predict what happens to the reaction rate at 5°C and at 80°C, and explain your reasoning using particle motion and denaturation.

3) A student says: "Because a catalyst speeds up the forward reaction, it must increase the amount of product formed." Write a two-sentence rebuttal that uses the ideas of forward/reverse rates and equilibrium.

Chemistry 30
Unit II — Thermochemistry
Lesson 8 — Empirical Effects of Catalysis
Part E — Catalysts in Industry & the Environment

🧠 Big Idea

A catalyst is a practical solution to a real problem: many reactions that are thermodynamically possible are simply too slow to be useful.

By lowering the activation energy (Ea), catalysts let reactions happen faster and more safely—

often at lower temperatures and pressures—without changing the overall ΔH or the equilibrium position. Catalysts change the clock, not the score.

5) Catalysis in Industry & the Environment: Evidence, Efficiency, and Trade-offs

When chemists and engineers talk about catalysts, they are usually thinking about one question:
“How can we get the reaction we want to happen fast enough to be useful—without making the process dangerous or wasteful?”

In an industrial setting, speed is not just a convenience. If a reaction is too slow, it forces a factory to use extreme conditions (very high temperature or pressure) to get a reasonable production rate. Those extreme conditions raise costs, increase safety risks, and often increase environmental impact.

A catalyst helps by providing an alternate reaction pathway with a lower Ea. That means a larger fraction of particle collisions have enough energy to react at the same temperature. The result is a faster reaction rate, even though the reactants and products (and the overall energy change, ΔH) are unchanged.

Real-world example: the Haber process (making ammonia, NH_3) uses an iron-based catalyst. The reaction is exothermic, so lower temperatures favour product at equilibrium—but lower temperatures also slow the reaction drastically. A catalyst is the compromise: it speeds the reaction so it can run at temperatures that are practical while still producing a useful yield.

Environmental example: catalytic converters in vehicles use catalysts (often platinum, palladium, and rhodium) to speed up reactions that convert harmful gases (like CO and NO_x) into less harmful products (CO_2 and N_2). The key idea is that the catalyst makes these conversions happen quickly at exhaust-pipe temperatures.

What does ‘evidence of catalysis’ look like in data?
• If you plot product formed versus time, the catalyzed reaction has a steeper slope at the start (greater initial rate).
• The catalyzed reaction often reaches the same final plateau faster. That plateau is set by thermodynamics (equilibrium), not by the catalyst.
• If the reaction goes to completion, both trials can produce the same total amount of product—the catalyst just gets you there sooner.

A careful student also watches for trade-offs:

Catalysts can be poisoned (blocked by impurities), they can be expensive, and they can be highly selective (great when you want one product; problematic when conditions change). In real systems, the 'best catalyst' is not just the fastest—it is the one that is fast, durable, selective, and economically realistic.

📘 Key Vocabulary

heterogeneous catalyst — catalyst in a different phase than the reactants (often a solid surface with gases or liquids).
homogeneous catalyst — catalyst in the same phase as the reactants (all gases or all aqueous).
adsorption — reactant particles stick to a catalyst surface (not the same as absorption).
active site — a specific region on a catalyst where reactants bind and react more easily.
selectivity — how strongly a catalyst favours one product pathway over others.
catalyst poison — a substance that reduces catalyst activity by blocking active sites (e.g., sulfur compounds).
catalytic converter — a device that uses catalysts to speed up reactions that reduce harmful exhaust gases.
turnover — the ability of a catalyst to participate repeatedly in reactions without being consumed overall.

⚠️ Common Error

Common confusion: "If the catalyst makes the reaction happen, it must change ΔH or make the reaction more exothermic."

Reality: ΔH depends only on the energy difference between reactants and products. A catalyst changes the pathway (Ea), not the starting and ending energy levels.

Another common error is mixing up rate with yield. A catalyst can make equilibrium happen faster, but it does not move the equilibrium position or change K. If a question shows two curves reaching the same plateau, the catalyst is the curve that reaches the plateau sooner—not the one with the higher plateau.

🎓 Diploma Tip

Diploma questions often use graphs or energy diagrams to test whether you can separate kinetics from thermodynamics.

If you see a product-vs-time graph:
• Catalyst present → steeper initial slope and shorter time to reach the plateau.
• Same plateau → equilibrium/maximum yield is unchanged.

If you see an energy diagram:
• Catalyst present → lower Ea peak(s).
• Reactant and product energy levels stay the same → ΔH is unchanged.

Also remember: a catalyst lowers Ea for both the forward and reverse reactions. That is why equilibrium is reached faster in either direction.

⏸ Pause & Think

1) A reaction reaches equilibrium in 2 minutes with a catalyst and 10 minutes without one. What does this tell you about the equilibrium position? What does it tell you about Ea?

2) On an energy diagram, how would you show (a) Ea, (b) ΔH, and (c) the effect of adding a catalyst?

3) In a factory, why might a catalyst be chosen even if it is expensive?

Chemistry 30 | Unit II | Lesson 8: Empirical Effects of Catalysis | Part F — Guided Practice + Diploma-Style Questions + Full Worked Solutions

🧠 Big Idea

A catalyst changes how quickly a reaction occurs—not how much energy the reaction ultimately releases or absorbs. Catalysts work by providing an alternate pathway with a lower activation energy (Ea). That makes it easier for reacting particles to succeed in forming products, so the rate increases. However, because the catalyst does not change the initial or final states of the reaction, the overall enthalpy change (ΔH) stays the same.

🧪 6) Guided Practice: Evidence, Energy Diagrams, and Reasoning

In this final part of the module, you will practice explaining catalysis the way diploma questions expect: using evidence, clear diagram labels, and careful language about what changes (Ea and rate) versus what does not change (ΔH and the energy difference between reactants and products). For each question, read the prompt slowly, decide what the question is truly asking, and then justify your answer using both words and energy-diagram logic.

🔑 Key Vocabulary

Catalyst: a substance that increases reaction rate by providing an alternate pathway with a lower activation energy, and is regenerated by the end.
Activation energy (Ea): the minimum energy barrier that must be overcome for reactants to reach the transition state.
Reaction coordinate (reaction progress): the pathway from reactants to products shown on an energy diagram.
Heterogeneous catalyst: catalyst in a different phase from reactants (often a solid surface).
Homogeneous catalyst: catalyst in the same phase as reactants.
Enzyme: a biological catalyst (usually a protein) that speeds reactions by stabilizing the transition state.
Denaturation: loss of an enzyme's shape (and therefore catalytic function) due to temperature or pH changes.

⚠️ Common Error

Saying “a catalyst makes the reaction more exothermic” or “a catalyst changes ΔH.”

A catalyst can make a reaction happen faster, so the temperature of the surroundings might change more quickly in a lab. That can feel like “more energy,” but it is not. The total energy transferred depends on the difference between reactants and products (ΔH), and that difference does not change when you add a catalyst.

🎓 Diploma Tip

When you explain catalysis on a written-response question, aim for three sentences that do three different jobs:
1) State what changes: “A catalyst provides an alternate pathway with a lower activation energy, so the reaction rate increases.”
2) State what does not change: “Because the initial and final states are unchanged, ΔH stays the same.”
3) Connect to the diagram: “On an energy diagram the catalyzed curve has a lower peak (Ea), but the vertical difference between reactants and products (ΔH) is identical.”

If you can hit all three, you are speaking diploma language.

⏸ Pause & Think

A student adds a catalyst to a reaction and says, “It released more energy because it got hotter.”

Before you read any solutions, write a one-paragraph response that corrects the claim using the words: system, surroundings, Ea, and ΔH.

📝 Practice Question 1

A reaction has ΔH = −85 kJ. On an energy diagram (potential energy vs reaction progress), the uncatalyzed pathway has Ea = 120 kJ.

a) Sketch the uncatalyzed energy diagram and label reactants, products, Ea, and ΔH.
b) A catalyst lowers the activation energy by 45 kJ. On the same axes, sketch the catalyzed

pathway and label the new Ea.
c) Explain, in words, why the catalyzed curve has a lower peak but the same ΔH.

✅ Worked Solution 1

a) Because ΔH is negative, the products are lower in potential energy than the reactants. Draw reactants at a higher level and products lower. The peak above reactants represents the transition state. Label Ea as the vertical rise from reactants to the peak (120 kJ). Label ΔH as the vertical drop from reactants to products (−85 kJ).

b) If the catalyst lowers Ea by 45 kJ, the new activation energy is:
Ea(catalyzed) = 120 kJ − 45 kJ = 75 kJ.
On the same axes, draw a second curve with a lower peak, but starting and ending at the same reactant and product levels.

c) Ea is the barrier height (how hard it is to reach the transition state). A catalyst provides an alternate pathway and stabilizes the transition state, so the barrier is lower. ΔH depends only on the energies of reactants and products (initial and final states). A catalyst does not change those states, so ΔH stays the same.

📝 Practice Question 2

You observe the decomposition of hydrogen peroxide:
$2\ H_2O_2(aq) \rightarrow 2\ H_2O(l) + O_2(g)$

Trial 1: No catalyst. Time to collect 50.0 mL of $O_2(g)$ = 180 s.
Trial 2: With $MnO_2(s)$. Time to collect 50.0 mL of $O_2(g)$ = 30 s.

a) Using the data above, compare the average rate of O_2 production in each trial.
b) Give two pieces of experimental evidence that support the claim “MnO_2 is acting as a catalyst.”
c) Explain why the same overall ΔH is expected for both trials, even though one is much faster.

✅ Worked Solution 2

a) For the same amount of product (50.0 mL O_2), a shorter time means a higher average rate.
Average rate ∝ (amount of O_2 produced) / time.
Trial 1 rate ∝ 50.0 mL / 180 s = 0.278 $mL{\cdot}s^{-1}$.
Trial 2 rate ∝ 50.0 mL / 30 s = 1.67 $mL{\cdot}s^{-1}$.
The catalyzed trial is about 1.67 / 0.278 ≈ 6.0 times faster.

b) Evidence:
• The rate increases dramatically when MnO_2 is present (cause-and-effect).
• MnO_2 is not consumed overall: you can recover the solid at the end (mass/appearance essentially unchanged), showing it is regenerated.
(Also acceptable: products are the same—only the rate changes.)

c) ΔH is determined by the energy difference between reactants and products. Both trials start with H_2O_2 and end with $H_2O + O_2$, so initial and final states are identical. The catalyst changes the pathway (Ea), not the start/end energies, so net ΔH is the same.

📝 Practice Question 3

A student draws an energy diagram for an exothermic reaction and labels the vertical drop from the peak down to the products as "activation energy."

a) Explain precisely what activation energy is, and where it is measured on an energy diagram.
b) Identify the quantity the student actually measured when they drew from the peak to the products.
c) Rewrite the student's explanation so it is correct and uses proper terms.

✅ Worked Solution 3

a) Activation energy (Ea) is the minimum energy barrier reactants must overcome to reach the transition state. On an energy diagram, Ea is measured from the reactants' energy level up to the peak.

b) The drop from the peak to the products corresponds to the activation energy for the reverse reaction (Ea,reverse): the barrier products must overcome to return to reactants.

c) A corrected explanation:
"The activation energy for the forward reaction is the energy difference between reactants and the transition state. In an exothermic reaction, products are lower than reactants, so the reverse reaction has its own activation energy measured from products to the transition state. A catalyst lowers both activation energies, but does not change ΔH."

📝 Practice Question 4

On a single set of axes, you sketch two curves: one uncatalyzed and one catalyzed. You correctly show that the catalyzed curve has a lower peak.

Why does lowering Ea increase reaction rate? Your answer must:
• refer to particle collisions and energy distribution
• explain what “more successful collisions” means
• avoid claiming that particles have more energy overall.

✅ Worked Solution 4

At a given temperature, particles have a spread (distribution) of kinetic energies. Only collisions involving particles with enough energy to reach the transition state can lead to products.

Lowering Ea does not increase particle energy. It lowers the minimum threshold required for a collision to succeed. Therefore, a larger fraction of the existing collisions already have energy ≥ Ea.

That increases the number of successful collisions per second, so the reaction rate increases even though the average kinetic energy (temperature) is unchanged.

📝 Practice Question 5

Enzymes are catalysts. Consider catalase, an enzyme that speeds the breakdown of hydrogen peroxide in cells.

a) Explain why enzymes are considered catalysts (use alternate pathway and regeneration).
b) A student heats catalase to 95°C and the reaction rate drops to almost zero. Explain this result without claiming that Ea increased because the enzyme “used up energy.”
c) State clearly whether ΔH changes when catalase is present, and justify your answer.

✅ Worked Solution 5

a) Enzymes increase reaction rate by providing an alternate pathway with lower Ea. They participate in the mechanism (often by binding reactants at an active site) and are regenerated at the end, so they are not consumed overall.

b) High temperature can denature the enzyme (change its 3-D shape). If the active site loses its shape, reactants no longer bind effectively and the enzyme can no longer stabilize the transition state. The low-Ea pathway is lost, so the reaction becomes very slow.

c) ΔH does not change. With or without catalase, the reaction starts with H_2O_2 and ends with $H_2O + O_2$, so the initial and final states are the same. The enzyme changes Ea (pathway), not ΔH.

📝 Practice Question 6

Diploma-style written response:

A student claims: “Catalysts do not change ΔH, so catalysts do not matter for energy.”

Write a response that agrees with part of the claim but corrects the misunderstanding. Your response must:
• distinguish between ‘energy change’ and ‘energy rate of transfer’
• refer to at least one real-world application (industrial or biological)
• include an energy-diagram statement (Ea vs ΔH).

✅ Worked Solution 6

A strong response includes:

• True part: Catalysts do not change ΔH; the overall energy change is set by the difference between reactants and products.

• Correction: Catalysts matter because they change how fast that energy change happens. The same total energy can be released quickly or slowly, and that affects safety, usefulness, and whether a process is practical.

• Diagram link: A catalyst lowers the peak (Ea) on an energy diagram but leaves the reactant-to-product energy difference (ΔH) unchanged.

• Application: Catalytic converters speed the conversion of harmful exhaust gases; enzymes like catalase make essential reactions fast enough at body temperature.

Conclusion: Catalysts do not change the net energy involved, but they matter enormously by controlling rate.

📝 Practice Question 7

Final Checkpoint (bridge forward):

Explain how the ideas from this lesson (Ea, alternate pathways, collision success) prepare you to study kinetics more formally.

In your answer, make at least one prediction about what would happen to reaction rate if temperature increased, and explain your reasoning.

✅ Worked Solution 7

Kinetics focuses on what controls reaction rate. This lesson introduced the key mechanism: particles must overcome an energy barrier (E_a) and only some collisions succeed.

If temperature increases, average kinetic energy increases and the energy distribution shifts so that a larger fraction of particles have energy $\geq E_a$. That increases the number of successful collisions per second, so reaction rate increases.

Catalysts are another kinetics lever: instead of changing particle energies, they lower the threshold (E_a). Later kinetics work will quantify these ideas using data, graphs, and relationships such as Arrhenius-style reasoning.

Chemistry 30 | Unit II | Lesson 8: Empirical Effects of Catalysis — Practice Pack (Student Version: Questions Only)

🧠 Big Idea

A catalyst changes how quickly a reaction occurs—not how much energy the reaction ultimately releases or absorbs. Catalysts work by providing an alternate pathway with a lower activation energy (Ea). That makes it easier for reacting particles to succeed in forming products, so the rate increases. However, because the catalyst does not change the initial or final states of the reaction, the overall enthalpy change (ΔH) stays the same.

🧭 Guided Practice — How to Use This Pack

This practice pack is designed to strengthen your ability to reason about catalysts using evidence and energy diagrams.

Work in order. For written responses, answer in full sentences. For calculations, show all steps and units.

When you get stuck, return to three questions: What are the reactants and products? Where is the energy barrier (Ea)? What stays the same when a catalyst is added (ΔH)?

[Instructional narrative text goes here]

🔑 Key Vocabulary

Catalyst: a substance that increases reaction rate by providing an alternate pathway with a lower activation energy, and is regenerated by the end.
Activation energy (Ea): the minimum energy barrier that must be overcome for reactants to reach the transition state.
Reaction coordinate (reaction progress): the pathway from reactants to products

shown on an energy diagram.
Heterogeneous catalyst: catalyst in a different phase from reactants (often a solid surface).
Homogeneous catalyst: catalyst in the same phase as reactants.
Enzyme: a biological catalyst (usually a protein) that speeds reactions by stabilizing the transition state.
Denaturation: loss of an enzyme's shape (and therefore catalytic function) due to temperature or pH changes.

⚠️ Common Error

Saying "a catalyst makes the reaction more exothermic" or "a catalyst changes ΔH."

A catalyst can make a reaction happen faster, so the temperature of the surroundings might change more quickly in a lab. That can feel like "more energy," but it is not. The total energy transferred depends on the difference between reactants and products (ΔH), and that difference does not change when you add a catalyst.

🎓 Diploma Tip

When you explain catalysis on a written-response question, aim for three sentences that do three different jobs:
1) State what changes: "A catalyst provides an alternate pathway with a lower activation energy, so the reaction rate increases."
2) State what does not change: "Because the initial and final states are unchanged, ΔH stays the same."
3) Connect to the diagram: "On an energy diagram the catalyzed curve has a lower peak (Ea), but the vertical difference between reactants and products (ΔH) is identical."

If you can hit all three, you are speaking diploma language.

⏸️ Pause & Think

Before you compare answers with a classmate or check in with your teacher, answer this in one or two sentences:

If a catalyst makes a reaction happen faster, why does that NOT automatically mean "more energy was released"?

Practice Question 1

An uncatalyzed reaction has reactants at 40 kJ and products at 10 kJ on a potential energy diagram. The peak (activated complex) is at 110 kJ.

(a) Calculate ΔH for the reaction.

(b) Calculate the activation energy, Ea, for the uncatalyzed reaction.

(c) A catalyst provides an alternate pathway with a lower peak at 70 kJ. Calculate the catalyzed Ea.

(d) In one clear sentence, explain what changed and what stayed the same when the catalyst was added.

Practice Question 2

Multiple Choice — Which statement best explains why a catalyst does not change the overall ΔH of a reaction?

A. A catalyst adds energy to reactants so they start at a higher potential energy.

B. A catalyst changes the initial and final energies equally, so ΔH cancels out.

C. A catalyst changes the pathway (Ea) but not the energies of reactants and products, so ΔH stays the same.

D. A catalyst increases the number of collisions, which makes the reaction more exothermic.

Practice Question 3

In a lab, hydrogen peroxide decomposes to water and oxygen gas. Two trials use the same mass of H_2O_2 at the same temperature. Trial 2 adds a small amount of $MnO_2(s)$ as a catalyst.

(a) Describe two observations you would expect to be different between Trial 1 and Trial 2 if MnO_2 is acting as a catalyst.

(b) Describe one measurement you would expect to be the same (or essentially the same) in both trials, and explain how that supports the idea that ΔH is unchanged.

Practice Question 4

Multiple Choice — A student wants to increase reaction rate. They can either heat the reactants or add a catalyst. Which comparison is correct?

A. Heating lowers Ea; a catalyst increases average kinetic energy.

B. Heating increases average kinetic energy; a catalyst lowers Ea.

C. Heating and catalysts both lower Ea, but catalysts work better.

D. Heating and catalysts both increase ΔH, which is why rate increases.

Practice Question 5

Enzymes are catalysts in living systems. Consider an enzyme-catalyzed reaction in human cells that runs quickly at 37°C but slows dramatically at 60°C.

(a) Explain why the reaction rate drops at 60°C even though higher temperature usually increases collision energy.

(b) Connect your explanation to what the enzyme (catalyst) is doing to the reaction pathway.

Practice Question 6

Multiple Choice — A reaction is exothermic. A catalyst is added. Which statement must be true?

A. ΔH becomes less negative because the catalyst stabilizes the products.

B. ΔH becomes more negative because the catalyst releases energy.

C. ΔH stays the same, but Ea decreases (for both forward and reverse directions).

D. Ea stays the same, but ΔH becomes more negative.

Practice Question 7

Written Response — A student claims: “Catalysts shift equilibrium to the products because the products form faster.”

Explain why this claim is incorrect. In your explanation, use the ideas of forward rate, reverse rate, and what a catalyst does (and does not) change.

Practice Question 8

Multiple Choice — A catalyst is added to a reversible reaction at equilibrium.

Which outcome is most accurate?

A. The equilibrium constant increases because the catalyst makes products form more easily.

B. The system shifts toward products because the forward reaction speeds up first.

C. The equilibrium position does not change, but equilibrium is reached faster because both forward and reverse rates increase.

D. ΔH becomes smaller because the catalyst lowers the energy change of the reaction.

Practice Question 9

A catalyst often participates in intermediate steps but is regenerated by the end.

Consider the two-step mechanism:

Step 1: A + Cat → A–Cat (intermediate)

Step 2: A–Cat + B → AB + Cat

(a) Identify the catalyst and the intermediate in this mechanism.

(b) Explain why the catalyst is not written in the net equation even though it matters for rate.

Practice Question 10

Energy Diagram Interpretation — A reaction profile has reactants lower than products, and the peak is very high.

(a) Decide whether the reaction is endothermic or exothermic, and justify using the diagram language (reactants vs products).

(b) Explain why the reaction could still be very slow at room temperature even if it is energetically “allowed.”

(c) State one change (other than adding a catalyst) that could increase the reaction rate, and explain using particle motion / collisions.

📝 Practice Question 11

Multiple Choice — Two experiments produce the same amount of product after a long time. In Trial 2 a catalyst is used.

Which graph best matches what you should expect for product vs time?

A. Trial 2 starts slower but ends higher.

B. Trial 2 rises faster at first but levels off at the same final amount.

C. Trial 2 rises at the same speed but levels off sooner at a higher final amount.

D. Trial 2 rises faster and levels off at a higher final amount because ΔH changed.

📝 Practice Question 12

Diploma-Style Written Response (Section C style) — You are given this claim:

“Because catalysts lower activation energy, they reduce the energy involved in the reaction.”

Write a clear, evidence-based response that:

• distinguishes Ea from ΔH

• references an energy diagram (describe what would be the same and what would change)

• uses at least one real-world example (e.g., catalytic converters or enzymes) to support your explanation.

Chemistry 30 — Unit II
Lesson 8: Empirical Effects of Catalysis
Solutions Pack

TEACHER SOLUTIONS (separate from Student Practice Pack)

✅ Solution 1 — Reading the Energy Diagram (Ea, ΔH, forward vs reverse)

Given: Reactants = 40 kJ, Products = 10 kJ, Peak (activated complex) = 110 kJ.

Step 1: Forward activation energy (Ea, forward) is the energy gap from reactants up to the peak:
$Ea_{forward} = 110 - 40 = 70$ kJ.

Step 2: Reverse activation energy (Ea, reverse) is the energy gap from products up to the peak:
$Ea_{reverse} = 110 - 10 = 100$ kJ.

Step 3: Enthalpy change (ΔH) compares products to reactants:
$\Delta H = 10 - 40 = -30$ kJ.

Interpretation: ΔH is negative, so the reaction is exothermic (the system loses energy; the surroundings gain it).

✅ Solution 2 — Multiple Choice (Why a catalyst does not change ΔH)

Correct answer: C.

Explanation: ΔH depends only on the energy of reactants and products (initial and final states). A catalyst provides an alternate pathway with a lower activation energy, so the peak is lower, but the starting and ending energy levels do not move. Therefore, ΔH is unchanged.

✅ Solution 3 — Multiple Choice (Catalyst affects rate; equilibrium unchanged)

Correct answer: B.

Explanation: A catalyst increases the rate of both the forward and reverse processes, so equilibrium is reached faster. However, it does not change the equilibrium position because it does not change ΔH or the relative energies of reactants and products.

✅ Solution 4 — Multiple Choice (Energy diagram with and without a catalyst)

Correct answer: D.

Explanation: The correct diagram must show the same reactant and product energy levels (so ΔH is unchanged), but a lower peak for the catalyzed pathway (lower Ea). Option D matches that pattern.

✅ Solution 5 — Multiple Choice (Product vs time graph: catalyzed vs uncatalyzed)

Correct answer: B.

Explanation: A catalyst changes how fast products form, not how much product exists at completion. So the catalyzed curve rises more steeply at first but levels off at the same final amount as the uncatalyzed curve.

✅ Solution 6 — Multiple Choice (What evidence supports 'catalyst lowers Ea'?)

Correct answer: A.

Explanation: Faster reaction at the same temperature is consistent with a lower activation energy barrier. Options B and D falsely suggest ΔH changes, and option C confuses rate with equilibrium yield.

✅ Solution 7 — Written Response (Catalyst: same yield, faster completion — explain)

A catalyst provides an alternate reaction pathway with a lower activation energy. That means a larger fraction of collisions have enough energy to reach the activated complex, so the reaction proceeds faster.

However, a catalyst does not change the energies of the reactants or products. Because the final state is the same, the net enthalpy change (ΔH) and the equilibrium position are unchanged. The reaction simply reaches the same endpoint more quickly.

Key wording to listen for: 'lower Ea', 'alternate pathway', and 'ΔH unchanged because reactant/product levels unchanged'.

✅ Solution 8 — Multiple Choice (Catalyst vs ΔH: choose the accurate statement)

Correct answer: C.

Explanation: A catalyst does not supply energy or change the overall energy released/absorbed. It lowers the barrier (Ea) so the reaction can proceed more readily. ΔH remains a property of the initial and final states.

✅ Solution 9 — Multiple Choice (Enzymes as catalysts: what changes?)

Correct answer: B.

Explanation: Enzymes (like all catalysts) increase reaction rate by lowering Ea, often by stabilizing the transition state and bringing reactants into an effective orientation. They are not consumed in the reaction, and they do not change ΔH.

✅ Solution 10 — Written Response (Compare catalyzed vs uncatalyzed profiles; compute Ea and ΔH)

Given: Catalyzed peak = 80 kJ; Uncatalyzed peak = 100 kJ; Reactants = 40 kJ; Products = 20 kJ.

Step 1: Activation energy (uncatalyzed):
$Ea_{uncat} = 100 - 40 = 60$ kJ.

Step 2: Activation energy (catalyzed):
$Ea_{cat} = 80 - 40 = 40$ kJ.

Step 3: Enthalpy change (same for both pathways):
$\Delta H = 20 - 40 = -20$ kJ.

Interpretation: The reaction is exothermic (negative ΔH). The catalyst lowers Ea by 20 kJ but does not change ΔH.

✅ Solution 11 — Multiple Choice (Catalyst and the activated complex)

Correct answer: C.

Explanation: The activated complex is the highest-energy point along the reaction pathway. A catalyst changes the pathway so that this highest-energy point is lower, meaning fewer joules are required to reach it. The catalyst does not ‘add’ energy to the system.

✅ Solution 12 — Diploma-Style Response (Design + interpret an empirical catalysis investigation)

A strong answer should include three components: evidence, energy-diagram reasoning, and the ΔH claim.

1) Two clear pieces of evidence for catalysis (rate evidence):
• The reaction occurs noticeably faster (e.g., faster gas production/bubbling, faster colour change, shorter completion time).
• A quantitative comparison (e.g., time-to-finish, rate = Δ(product)/Δt, or a steeper initial slope on a graph).

2) What the energy diagram must show:
• Reactants and products at the same energy levels as the uncatalyzed reaction (so ΔH is unchanged).
• A lower peak for the catalyzed pathway (lower activation energy).

3) How to conclude 'ΔH is unchanged' using words:
Because ΔH compares only the energy of reactants and products, and a catalyst does not change those levels, the net energy released/absorbed is the same. The catalyst only changes the pathway (E_a), not the start or end points.

Optional extension students may add: Enzymes are catalysts in living systems; they work by binding substrates, orienting them, and stabilizing the transition state—speeding reactions that would otherwise be too slow at body temperature.

🧪 Chemistry 30 | Thermochemical Changes | Lesson 8

Empirical Effects of Catalysis

Exit Ticket • 1–2 diploma-style prompts

🌿 Big Idea

A catalyst changes *how fast* a reaction happens by lowering the activation energy (E_a), but it does not change *how much* energy is involved overall (ΔH) and it does not change the equilibrium position.

In other words: catalysts change the pathway, not the start and finish.

✅ Instructions

Answer in complete sentences. Use key terms correctly (system, surroundings, activation energy E_a, ΔH, catalyst, reaction profile).

Where asked, include a clearly-labelled energy diagram.

Aim for *reasoning + evidence*, not memorized statements.

📝 Exit Prompt 1 (Evidence + Diagram)

You run the same reaction twice at the same temperature.

• Trial A (no catalyst): bubbles appear slowly; the reaction takes several minutes.

• Trial B (with catalyst): vigorous bubbling; the reaction finishes quickly.

a) Use the ideas of activation energy and particle collisions to explain *why* Trial B is faster.

b) On a reaction-progress vs potential-energy diagram, sketch Trial A and Trial B on the *same axes*.

Label E_a (both), ΔH, and indicate clearly what changes and what does not.

c) One student claims: "Because the catalyzed reaction is faster, it must release more heat." Explain why this claim is incorrect.

🧭 Exit Prompt 2 (Bridge to Kinetics & Equilibrium)

An exothermic reversible reaction has reached equilibrium in a sealed container.

a) Predict what happens immediately after a catalyst is added: does the equilibrium position shift, or not? Explain.

b) Even if the equilibrium position does not change, what *does* the catalyst change about the system over time?

c) State one piece of evidence you would look for in a lab to support your claim in (b).

📋 Quick Rubric Notes + Looking Ahead

How your response is assessed (constructed response):

• Concept Accuracy: correct use of E_a, ΔH, catalyst, system vs surroundings, equilibrium language.

• Reasoning & Evidence: you explain *why* (cause-and-effect) and point to observations/diagrams as evidence.

• Representation: energy diagram is labelled, readable, and matches your explanation.

• Communication: clear sentences, correct scientific terms and units where relevant.

Looking ahead: the next unit builds on this by quantifying *rate* (kinetics) and explaining how systems reach equilibrium. A helpful habit: always separate “speed of reaction” from “energy released/absorbed.”